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Track 1

AI Systems & Workflow Innovation Challenge

Redrob Context & Ideathon Scope

Redrob AI is building an AI-native ecosystem spanning hiring, productivity, communication, discovery, workflows, recommendations, automation, and intelligent user experiences.

Participants are encouraged to think beyond standalone applications and explore how their ideas can:

Extend existing
Redrob capabilities

Improve
experiences for
Redrob users

Introduce
new AI-native
workflows

Strengthen ecosystem
engagement and
network effects

Important

Participants are not expected to build a completely new platform from scratch.

Strong submissions will demonstrate:

How the idea leverages existing Redrob capabilities

What new capability, workflow, or value layer is introduced

How the idea strengthens the overall Redrob ecosystem

Why Redrob is uniquely positioned to enable this experience

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Track 01

AI Systems & Workflow Innovation Challenge

Team Name : Hyperion

Problem Statement : Ideathon Challenge 1 – The AI Systems Architect: Reimagining Work

Team Members : Nitanshu Tak (Team Leader), Swati Dubey

Problem Definition

Questions

- Recruiter shortlists go stale the moment they're created. A candidate ranked #1 today may stop responding tomorrow – but the shortlist doesn't know that until someone re-runs the search.
- Every recruiter running an active search on Redrob, and every candidate whose real availability (response rate, recent activity) silently diverges from their static profile score.
- Search-and-rank is a one-time event triggered by a human. Nothing on the platform watches a shortlist after it's created, so good matches quietly decay into wasted outreach.

Opportunity & Vision

Questions

- Redrob already has the exact signals needed to solve this – 23 behavioral fields per candidate, updated continuously. Today they're inputs to a one-time score. Turning them into a live, ongoing signal is a capability extension, not a new product.
- A shortlist that stays accurate on its own. Recruiters open Redrob to a feed of meaningful changes – not a static list they have to remember to refresh.

Solution Overview

Questions

- Continuum: an always-on intelligence layer that continuously re-scores active candidate pools against open JDs as behavioral signals update, surfacing only meaningful rank changes with a live explanation – built directly on Prism's five-layer scoring architecture (our Track 1 submission).
- It initiates re-ranking itself and decides what's worth surfacing – the recruiter never has to ask. AI-assisted tools wait for a query; Continuum watches, decides, and notifies on its own.
- Builds directly on Redrob's existing candidate search/discovery surface and the same 23 redrob_signals fields already collected per profile – this is a new intelligence layer over current infrastructure, not a parallel system.

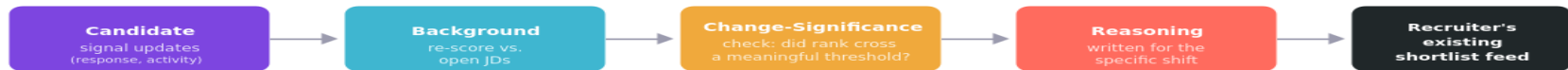
User Journey / Workflow Diagram

Questions

- Recruiters use their existing Redrob shortlist as normal – Continuum runs invisibly behind it, no new screen to learn.
- Signal updates → background re-score vs. open JDs → diff vs. last-notified rank → meaningful shifts surface with reasoning attached.
- Inside the existing search/shortlist surface – a live status indicator and notification feed, not a new destination.

Mandatory Visual

- Workflow Diagram



No new screen, no new workflow — the recruiter's existing Redrob shortlist simply stays current on its own.

AI Logic & Decision Flow

Questions

- On every signal update: a Re-Scoring Agent recomputes Prism's semantic + trust score for that candidate against their open JDs.
- A Change-Significance Agent checks if the shift crosses a meaningful threshold (e.g. enters/exits top 10) – noise is suppressed.
- Re-Scoring → Change-Significance → Explainability → Outreach-Drafting (only on top-rank entry) – a strict pipeline, not free-form chatter.

Mandatory Visual

- AI Flow Diagram



Strict linear pipeline — each agent has one job and passes a structured object forward. No free-form agent-to-agent chatter.

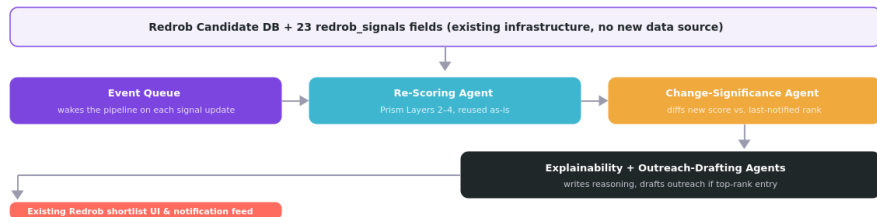
System Architecture

Questions

- Re-Scoring Agent (Prism Layers 2-4), Change-Significance Agent, Explainability Agent, Outreach Agent, and an event queue.
- A strict pipeline, not a chat loop – each agent has one job and passes a structured object to the next.
- Redrob's candidate database, the 23 redrob_signals fields, the search UI, and the notification system – no new data source.

Mandatory Visual

- Architecture Diagram



Data, Context & Intelligence Layer

Questions

- Exactly what Redrob already has: candidate profiles, career history, and the 23 redrob_signals fields.
- Last-notified score/rank is cached per candidate; new events are diffed against that cache to find real changes.
- Existing signals go from a one-time read to a continuous watch – same data, no added burden on anyone.

Mandatory Visual

- Data Flow Diagram



Same data Redrob already collects — the only new component is the cache + diff layer that turns a one-time read into a continuous signal.

Scalability & Technical Feasibility

Questions

- As a background job queue reacting to signal-update events, reusing Prism's existing CPU-only scoring code (Layers 2-4) — no new ML infrastructure, just a scheduler and a diff layer.
- Re-scoring is per-candidate, per-affected-JD only — not a full 100K re-rank on every event — so cost scales with actual activity volume, not total pool size.
- Notification fatigue if thresholds are too sensitive; needs careful tuning of the Change-Significance Agent's cutoff — the same kind of precision-vs-noise tradeoff we already validated in Prism's honeypot detection on real data.

Redrob Ecosystem Integration

Questions

- Candidate search, the 23 redrob_signals fields, the existing shortlist UI, and the recruiter notification system.
- Continuous, self-updating shortlists — a standing capability no current rank tool offers.
- Recruiters open Redrob more often, since it proactively has something new to show — turning search into a daily habit.
- The same pattern extends to candidates — alerting them to new matching roles, making Redrob two-sided.

Mandatory Visual

- Ecosystem Integration Diagram



One new intelligence layer touches every existing Redrob product surface — not a bolt-on, a connective layer.

Impact & Success Metrics

Questions

- Reduced time-to-contact (recruiters reach available candidates faster), fewer wasted outreach attempts to candidates who've gone quiet, and higher shortlist-to-hire conversion.
- Track notification-to-contact rate, shortlist staleness (time between rank change and recruiter action), and outreach response rate before/after adoption.
- Recruiters spend less time re-searching and more time talking to genuinely available candidates; Redrob gains a sticky, daily-use feature that strengthens platform engagement.

Future Roadmap

Questions

- Year 1: live re-ranking for recruiters. Year 2: symmetric alerts for candidates on new matching roles. Year 3: predictive signals (forecasting candidate availability shifts before they happen).
- Two-sided continuous matching, automated interview scheduling triggered by a fresh top-rank event, and a trust score that follows candidates across every role they're considered for.
- Redrob's own vision of an AI-native operating system for how India hires – not a tool you query, but a layer that's always working on your behalf.



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THANK YOU